MAR BASELIOS
COLLEGE OF ENGINEERING AND TECHNOLOGY
AUTONOMOUS

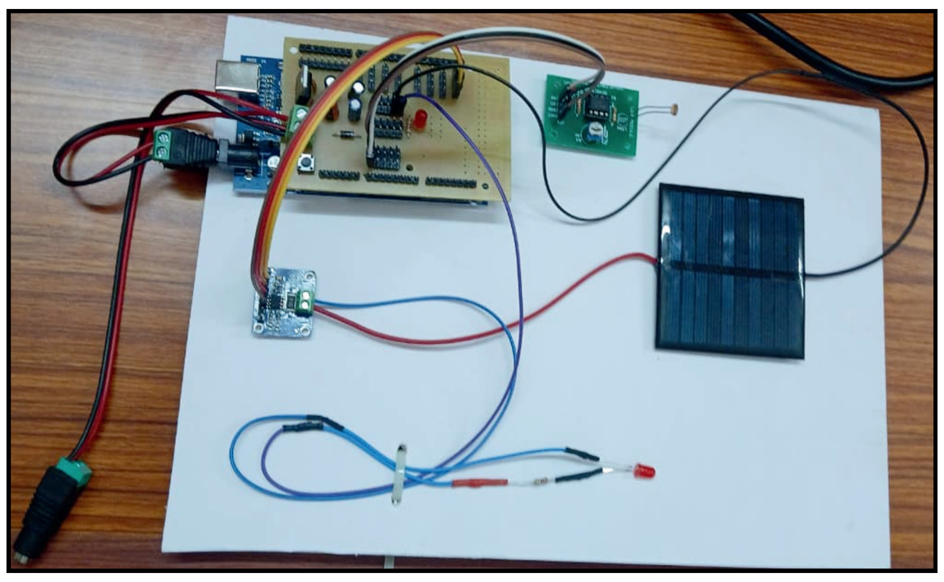
AI INTEGRATED SOLAR PERFORMANCE OPTIMIZATION

Solar energy is the leading carbon-free technology for power generation Kerala's solar capacity grew from 100 MW (2018) to over 1,700 MW (2025) Solar PV systems suffer from faults, degradation, shading & soiling issues Traditional monitoring is reactive — delayed fault detection causes major energy losses AI/ML-based proactive monitoring can significantly improve efficiency

OBJECTIVES

- Develop an intelligent real-time monitoring system for PV plants
- Implement ML models for anomaly detection (5-class fault classification)
- Design a predictive maintenance framework using Random Forest
- Build accurate solar power forecasting models
- Deploy interactive dashboards with MQTT live-stream integration
- Hardware Prototype

HARDWARE RESULTS



NORMAL
Voltage: 3.30V ·
Current: 6.2mA
Load Power: 21.0mW

SHADING
Voltage: 3.57V ·
Current: 7.3mA
Power drop detected

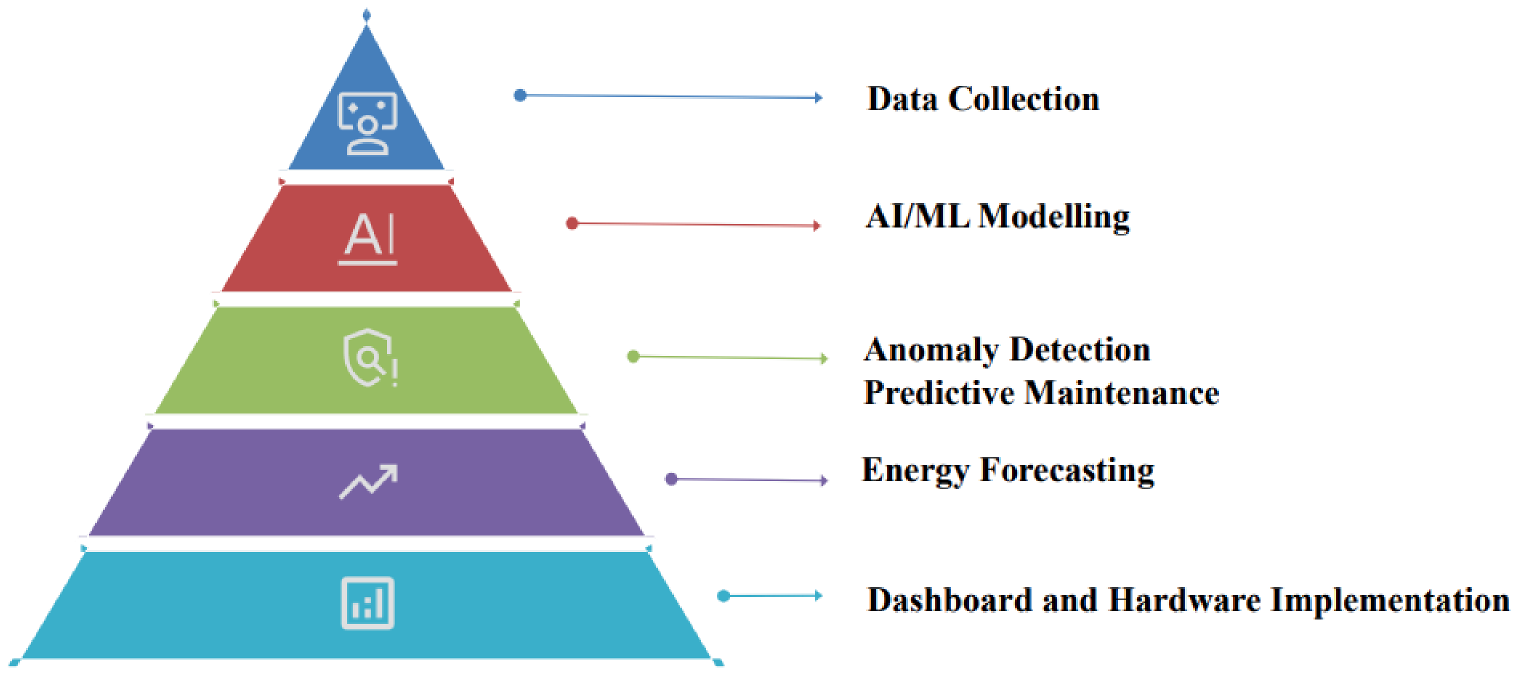
OPEN CIRCUIT
Voltage: 1.04V ·
Current: 0.5mA
Radiance: 0 (no light)

- Solar panel generates voltage
- Sensors measure current, voltage, light intensity
- Arduino processes data and sends to system
- Output visualized via dashboard

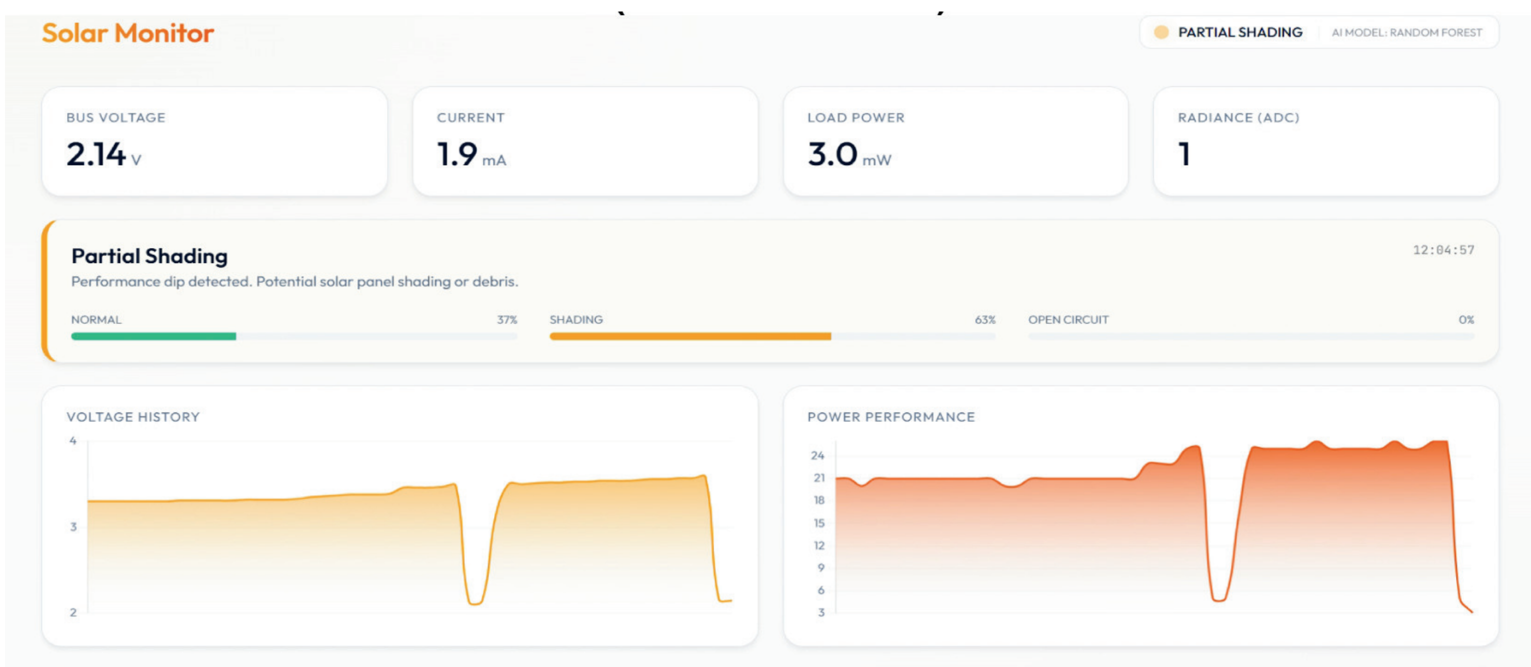
5 CLASS FAULT DETECTION

NORMAL | SHORT CIRCUIT | OPEN CIRCUIT | DEGRADATION | SHADING

- Detects 5 types of faults: Normal, Short Circuit, Open Circuit, Degradation, and Shading using PV system data.
- Trained on a large dataset (1M+ samples) with key features like voltage, current, irradiance, and panel temperature.
- Hybrid model achieves high accuracy across all fault classes, ensuring reliable and robust fault detection.



Hardware Implemetation Dashboard



Enhances solar PV efficiency through intelligent monitoring and fault detection.

Enables smart energy systems using AI-driven monitoring and automation.

Reduces carbon emissions by improving solar energy utilization and reliability.

Optimizes maintenance to reduce energy loss and operational waste.

Team Members :
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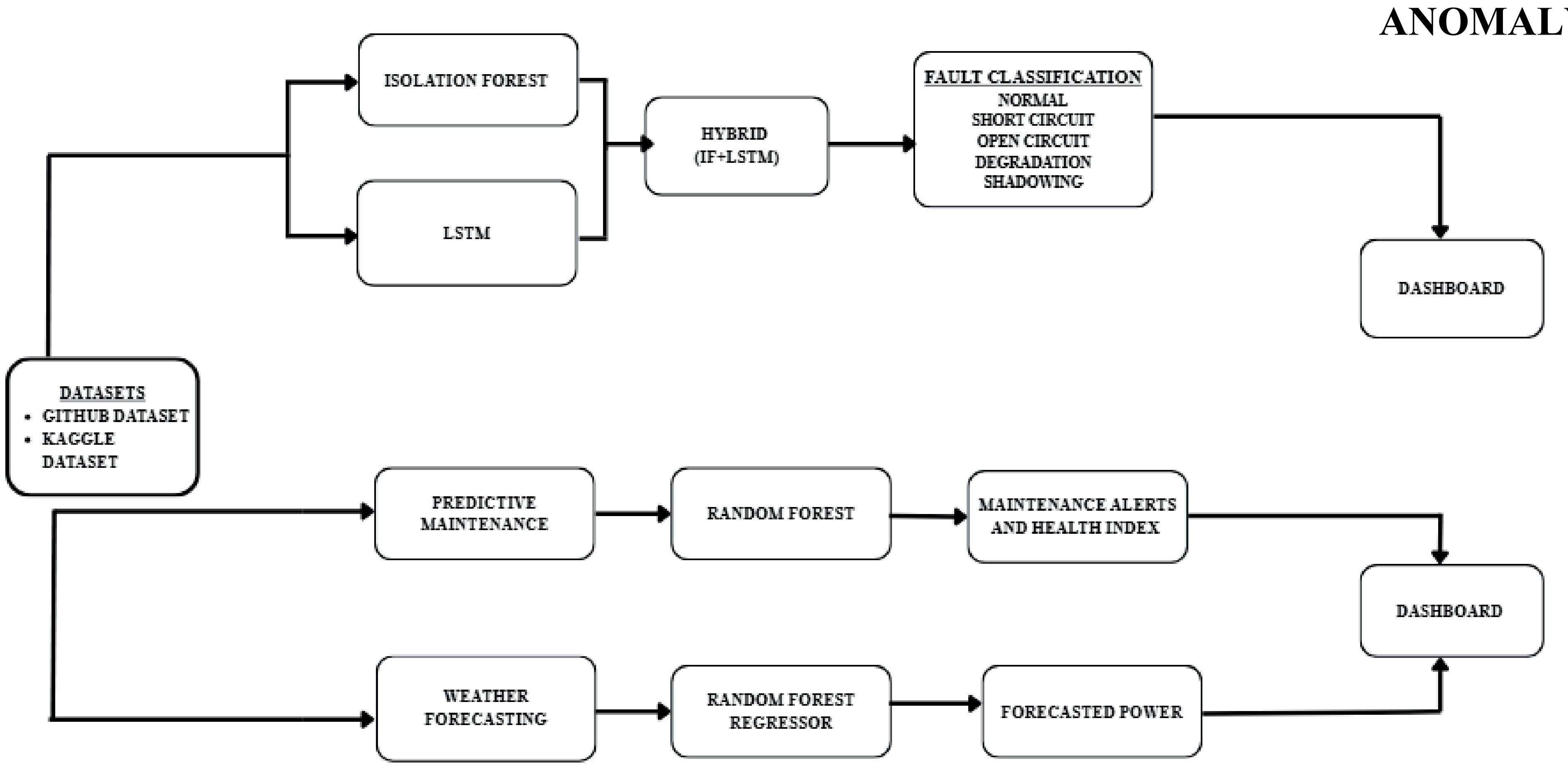
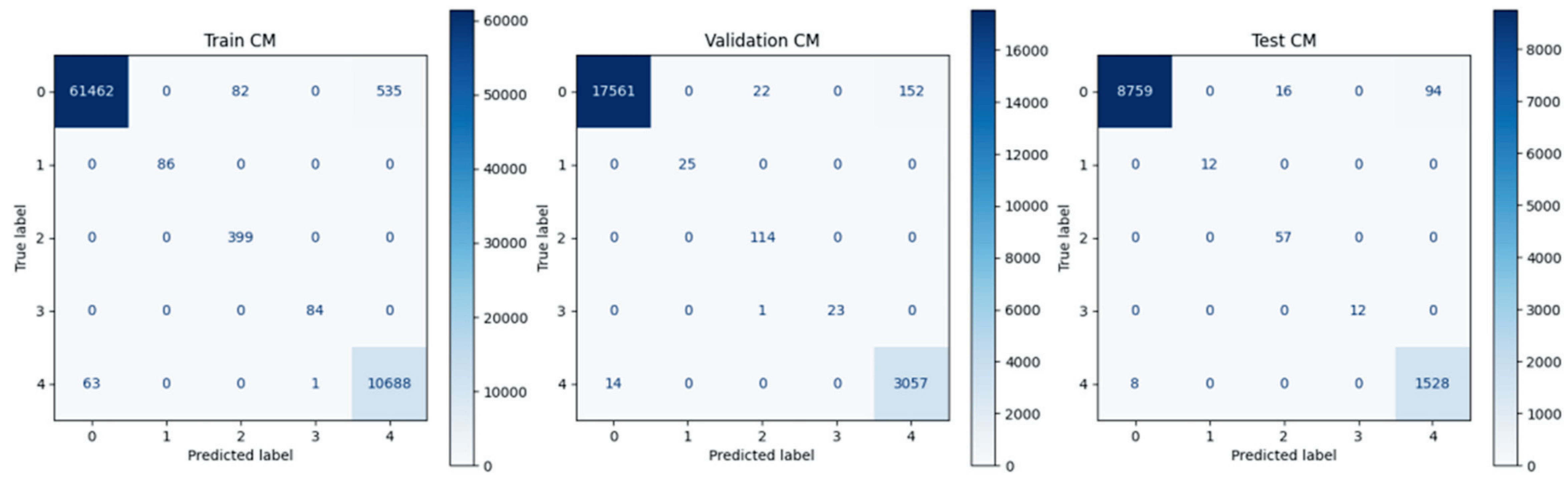
SOFTWARE METHODOLOGY



ANOMALY DETECTION-MODEL COMPARISON

MODEL	ACCURACY	F1
Isolation Forest	74%	0.19
LSTM	84%	0.67
HYBRID	98%	0.98

ANOMALY DETECTION RESULTS



PREDICTIVE MAINTENANCE-RANDOM FOREST

- Dataset: Kaggle Solar Generation + Weather (68,778 records)
- Parameters: AC/DC Power, Irradiation, Module Temp, Ambient Temp
- Predicts fault occurrence before failure — enables scheduled maintenance

94.8%	95.4%	94.9%	0.897
Training Accuracy	Validation Accuracy	Testing Accuracy	AUC-PR Score

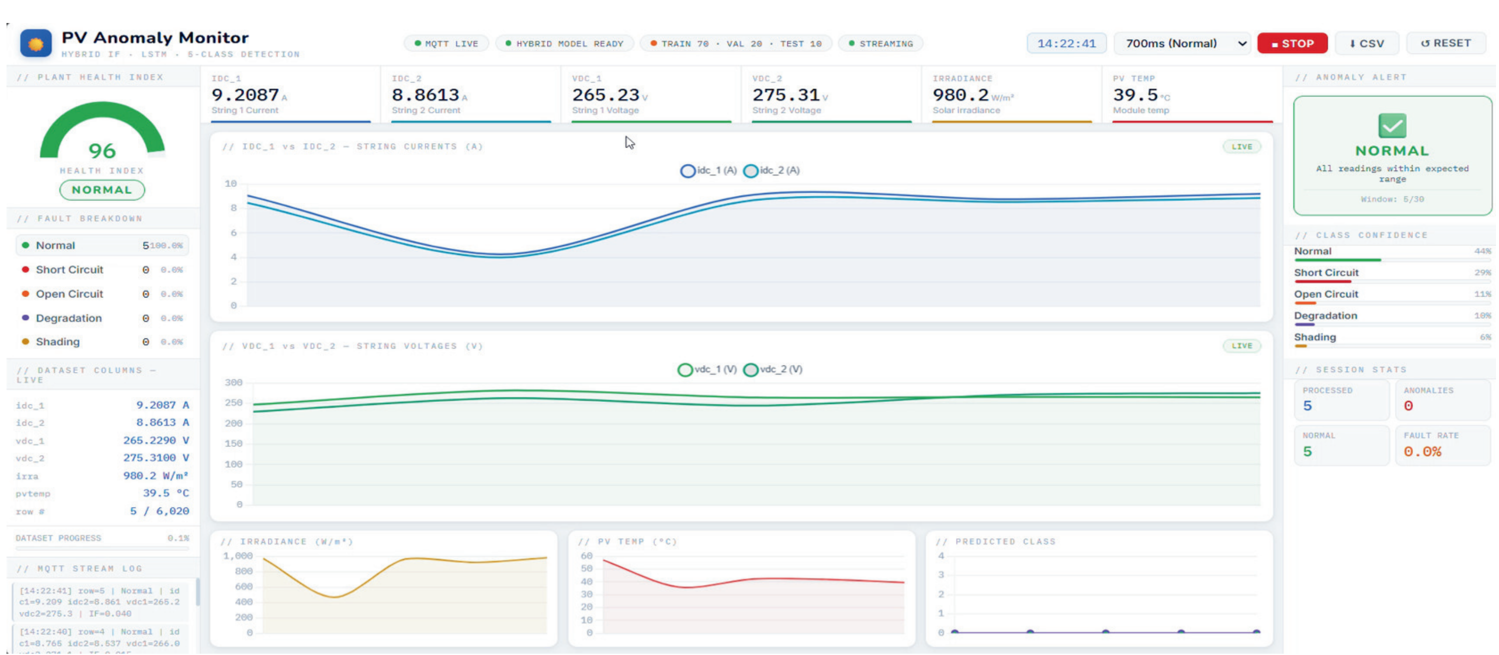
POWER FORECASTING-RANDOM FOREST REGRESSOR

- Actual vs. predicted power output shows near-perfect overlap across time
- Prediction errors clustered near zero — minor deviations due to weather fluctuations
- Supports grid planning, load balancing, and energy management

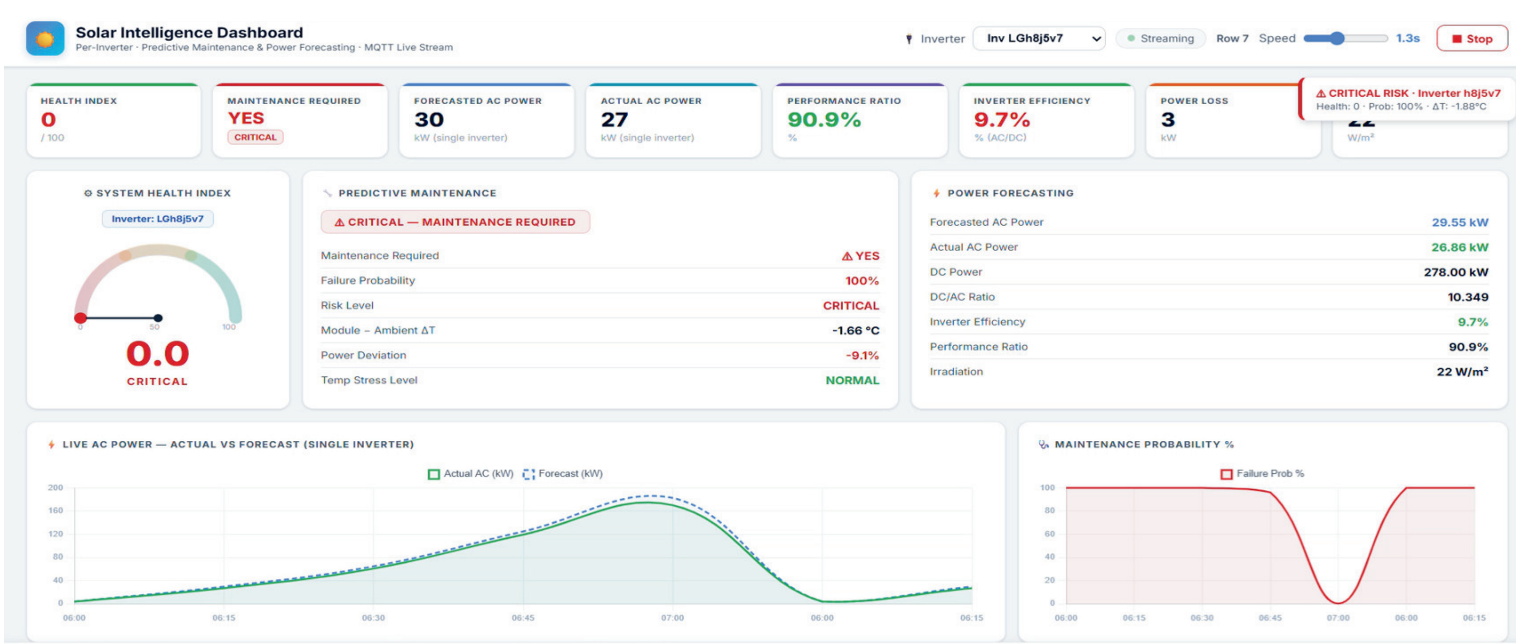
0.997	0.993	94.9%	0.897
Training R2	Validation Accuracy	Testing Accuracy	AUC-PR Score

DASHBOARD IMPLEMENTATION

Anomaly Detection Dashboard



Predictive Maintenance Dashboard



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